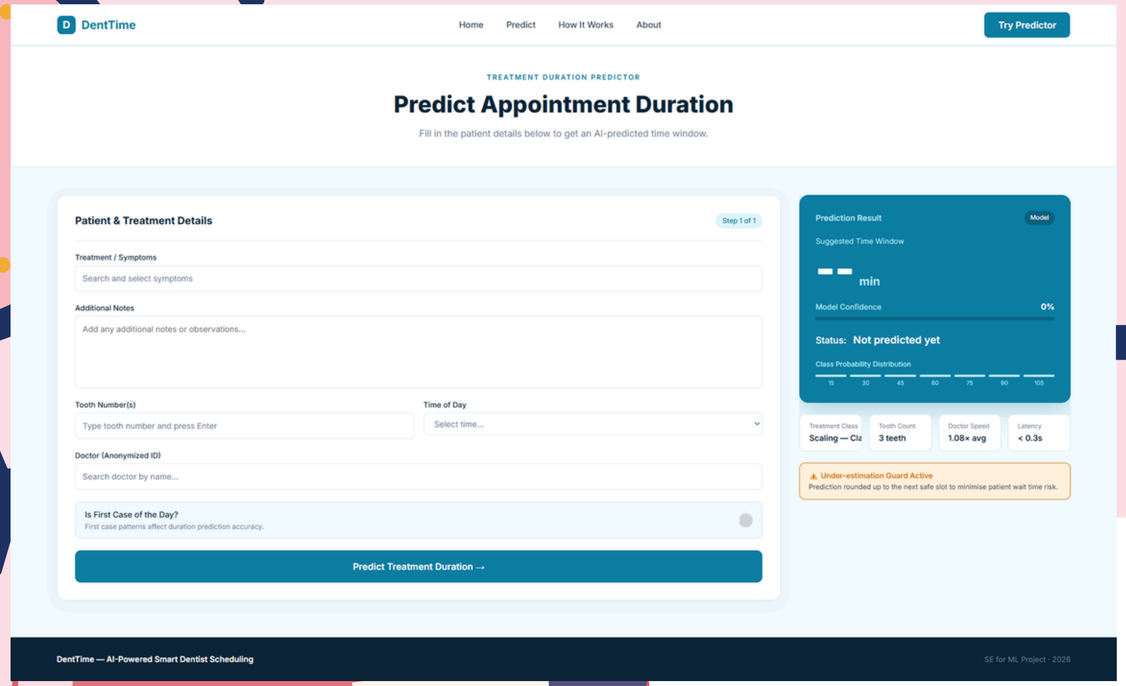
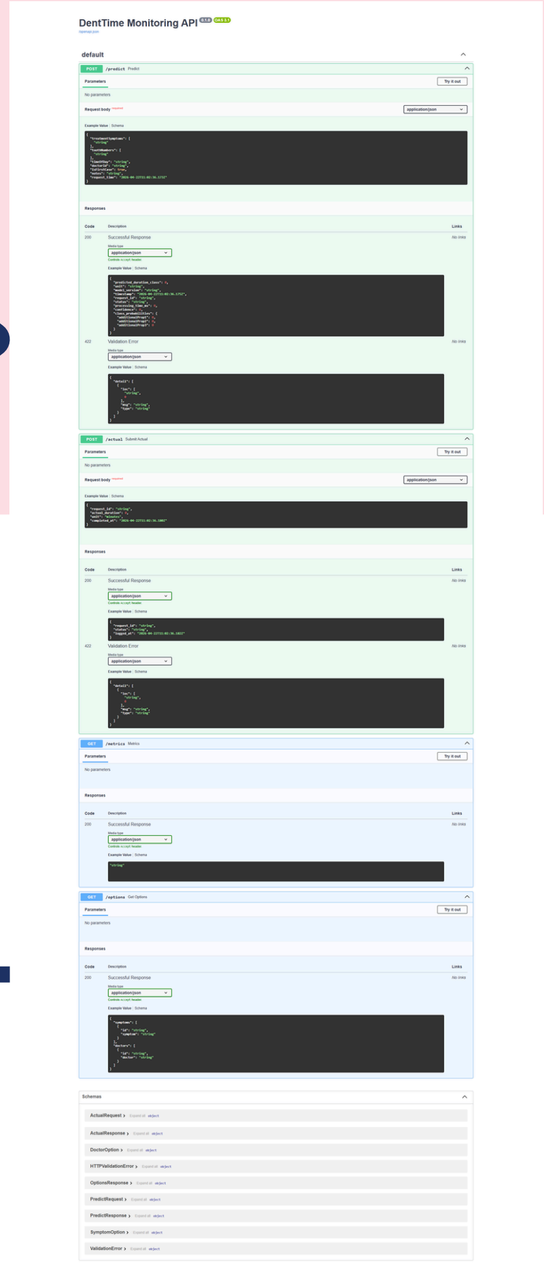
MONITORING & ALERTING OVERVIEW

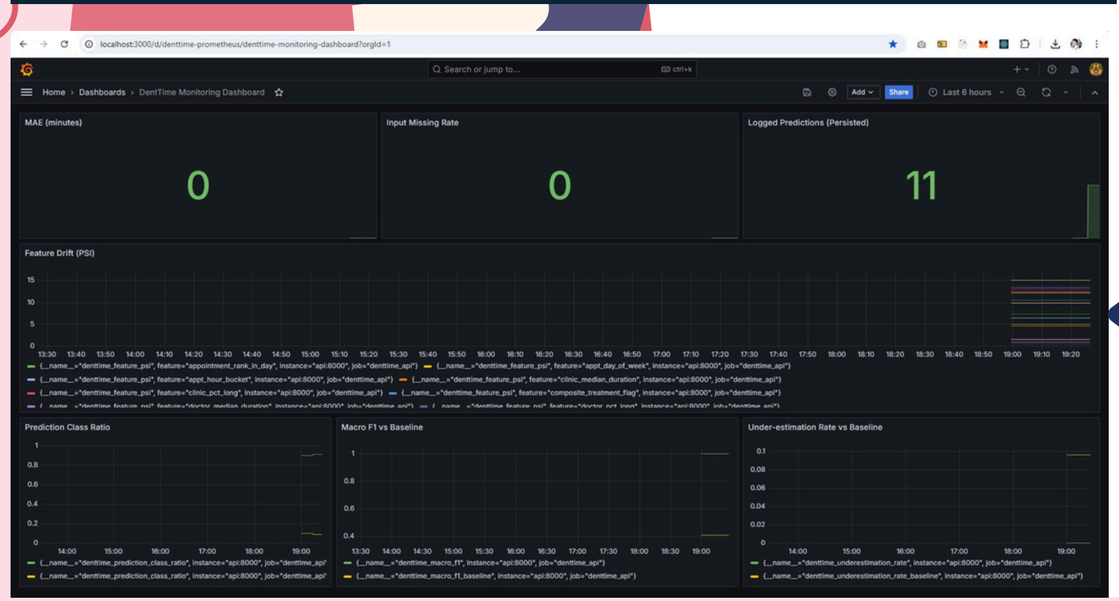
Goal: monitor input, output, and post-deployment model quality

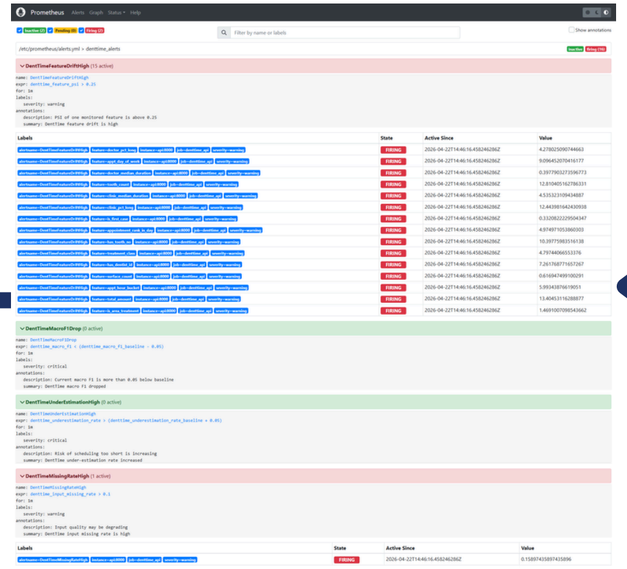
Stack: FastAPI + Prometheus + Grafana

Objective: detect drift, performance change, and degradation early







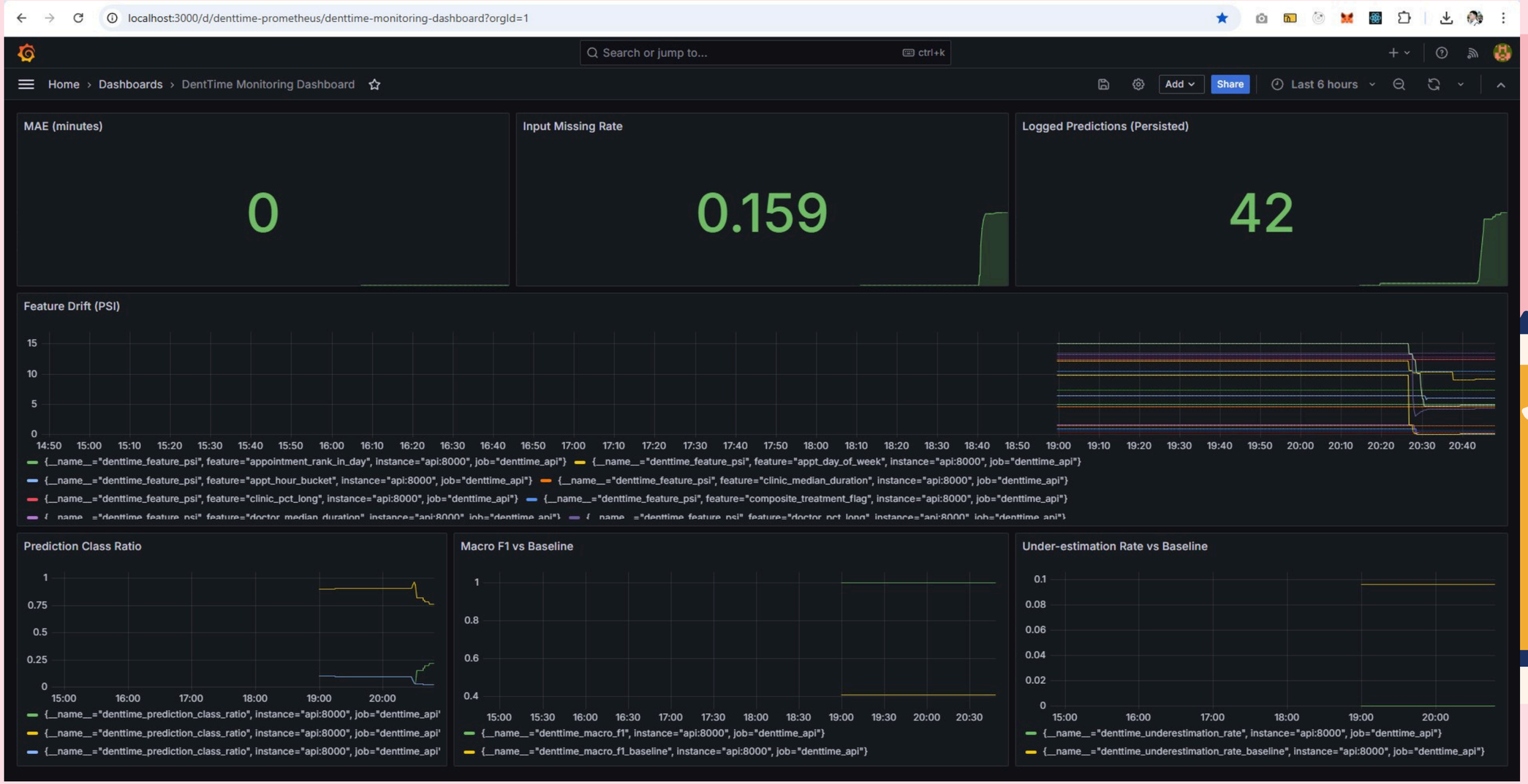


FRONTEND → API → PROMETHEUS → GRAFANA

WHAT EXACTLY IS MONITORED

1. Input monitoring
2. Input missing rate
3. Feature drift (PSI) vs baseline
4. Output monitoring
5. Prediction class ratio
6. Logged predictions
7. Performance monitoring
8. Macro F1 vs baseline
9. MAE (minutes)
10. Under-estimation rate vs baseline

WHAT EXACTLY IS MONITORED



GRAFANA DASHBOARD

MONITORING ARCHITECTURE AND DATA FLOW

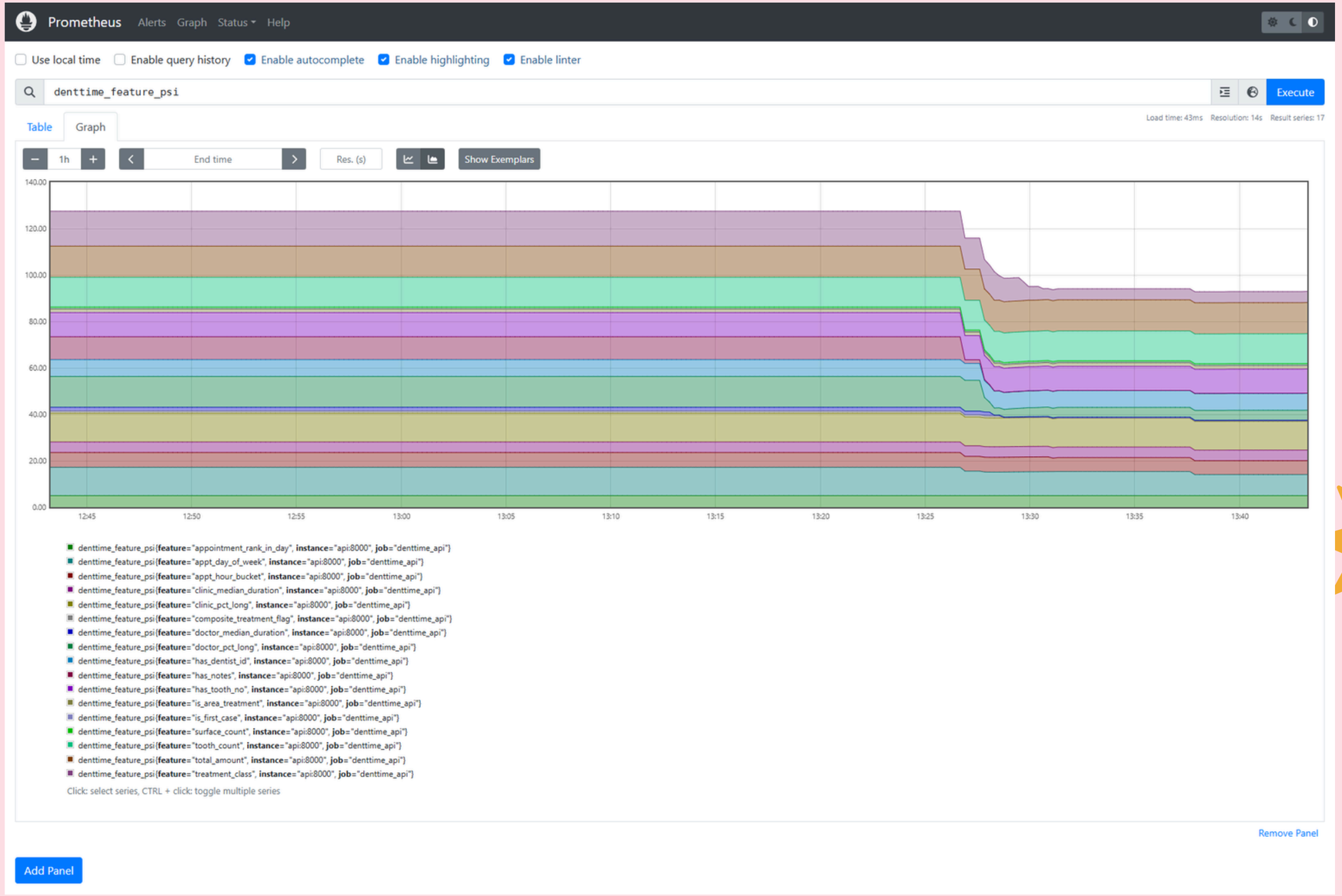
- /predict
  → log request + prediction
- /actual
  → log actual outcome
- monitoring/update_metrics.py
  → compute rolling metrics from persisted records
- Prometheus scrapes /metrics
- Grafana visualizes metrics
- Alerts are evaluated in Prometheus

User/API Request
→ FastAPI /predict
→ SQLite / persisted prediction log
→ FastAPI /actual
→ labeled predictions
→ update_metrics.py
→ /metrics
→ Prometheus
→ Grafana + Alerts

BASELINE AND DRIFT LOGIC

Baseline source: feature_stats.json
Drift monitored on 17 numeric features
Current target classes: 15, 30, 45, 60, 90, 75, 105
Example alert: denttime_feature_psi > 0.25
Purpose: detect distribution shift before silent failure

BASELINE AND DRIFT LOGIC



PROMETHEUS QUERY DENTTIME_FEATURE_PSI

BASELINE AND DRIFT LOGIC

Prometheus Alerts Graph Status Help

Inactive (2) Pending (0) Firing (2)

Filter by name or labels

Show annotations

/etc/prometheus/alerts.yml > denttime_alerts

Inactive Firing (16)

DentTimeFeatureDriftHigh (15 active)

```
name: DentTimeFeatureDriftHigh
expr: denttime_feature_psi > 0.25
for: 1m
labels:
  severity: warning
annotations:
  description: PSI of one monitored feature is above 0.25
  summary: DentTime feature drift is high
```

Labels	State	Active Since	Value
alertname=DentTimeFeatureDriftHigh feature=doctor_pct_long instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.278025090744663
alertname=DentTimeFeatureDriftHigh feature=appt_day_of_week instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	9.096452070416177
alertname=DentTimeFeatureDriftHigh feature=doctor_median_duration instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.3977903273596773
alertname=DentTimeFeatureDriftHigh feature=tooth_count instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	12.810405162786331
alertname=DentTimeFeatureDriftHigh feature=clinic_median_duration instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.535323109434887
alertname=DentTimeFeatureDriftHigh feature=clinic_pct_long instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	12.443981642430938
alertname=DentTimeFeatureDriftHigh feature=is_first_case instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.3320822229504347
alertname=DentTimeFeatureDriftHigh feature=appointment_rank_in_day instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.974971053860303
alertname=DentTimeFeatureDriftHigh feature=has_tooth_no instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	10.39775983516138
alertname=DentTimeFeatureDriftHigh feature=treatment_class instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.79744066553376
alertname=DentTimeFeatureDriftHigh feature=has_dentist_id instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	7.261768771657267
alertname=DentTimeFeatureDriftHigh feature=surface_count instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.616947499100291
alertname=DentTimeFeatureDriftHigh feature=appt_hour_bucket instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	5.99343876619051
alertname=DentTimeFeatureDriftHigh feature=total_amount instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	13.40453116288877
alertname=DentTimeFeatureDriftHigh feature=is_area_treatment instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	1.4691007098543662

DentTimeMacroF1Drop (0 active)

```
name: DentTimeMacroF1Drop
expr: denttime_macro_f1 < (denttime_macro_f1_baseline - 0.05)
for: 1m
labels:
  severity: critical
annotations:
  description: Current macro F1 is more than 0.05 below baseline
  summary: DentTime macro F1 dropped
```

DentTimeUnderEstimationHigh (0 active)

```
name: DentTimeUnderEstimationHigh
expr: denttime_underestimation_rate > (denttime_underestimation_rate_baseline + 0.05)
for: 1m
labels:
  severity: critical
annotations:
  description: Risk of scheduling too short is increasing
  summary: DentTime under-estimation rate increased
```

DentTimeMissingRateHigh (1 active)

```
name: DentTimeMissingRateHigh
expr: denttime_input_missing_rate > 0.1
for: 1m
labels:
  severity: warning
annotations:
  description: Input quality may be degrading
  summary: DentTime input missing rate is high
```

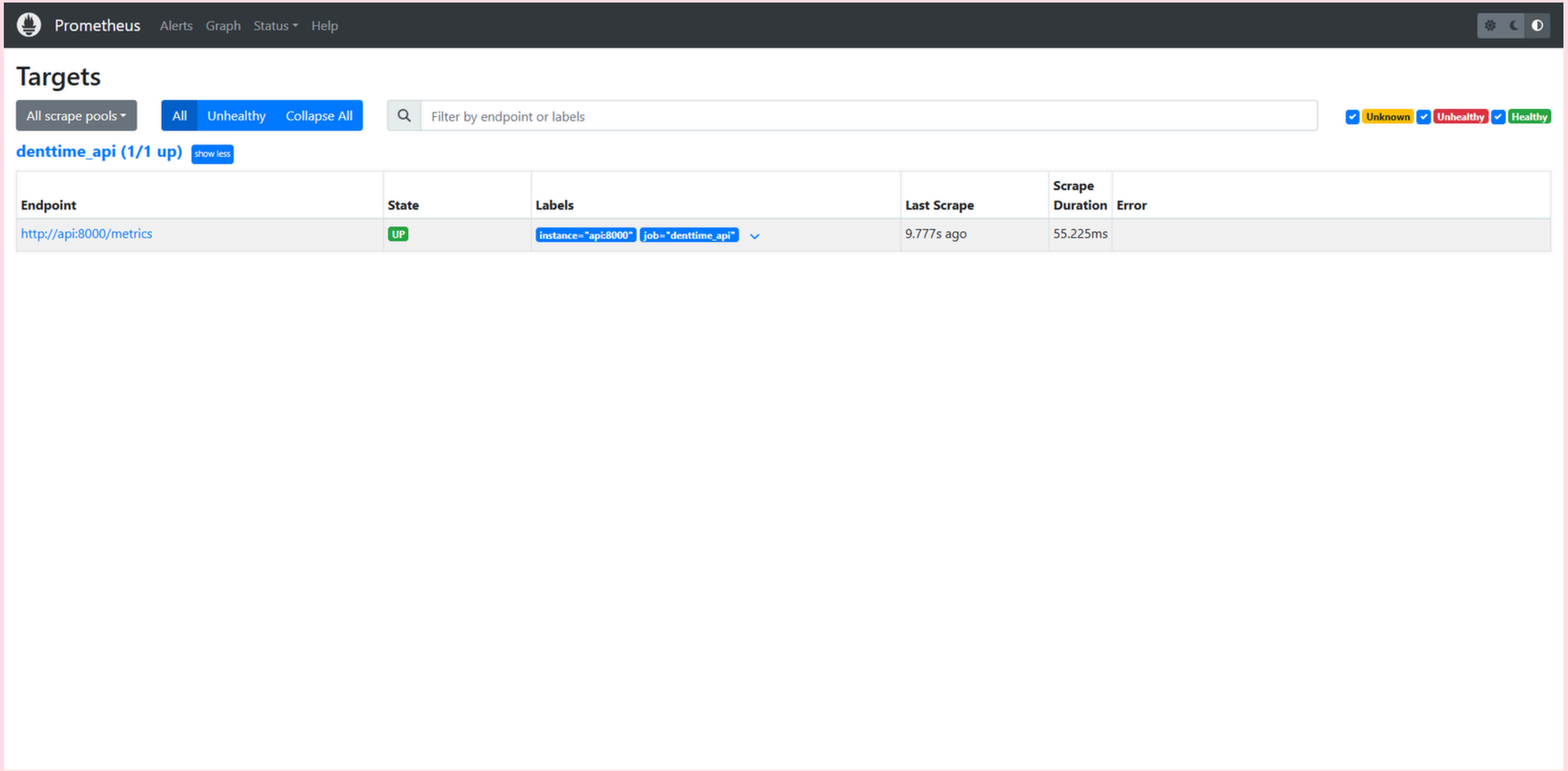
Labels	State	Active Since	Value
alertname=DentTimeMissingRateHigh instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.15897435897435896

ALERTS PAGE ที่ FIRING เรื่อง FEATURE DRIFT

ACTUAL EVIDENCE FROM THE WORKING SYSTEM

- Prometheus target = UP
- /predict → /actual → metrics smoke flow = success
- Grafana dashboard renders live metrics
- Alerts page shows firing state
- Monitoring is integrated end-to-end, not mock-only

ACTUAL EVIDENCE FROM THE WORKING SYSTEM



TARGETS = UP

ACTUAL EVIDENCE FROM THE WORKING SYSTEM

Prometheus Alerts Graph Status Help

Inactive (2) Pending (0) Firing (2)

Filter by name or labels

Show annotations

/etc/prometheus/alerts.yml > denttime_alerts

Inactive firing (16)

DentTimeFeatureDriftHigh (15 active)

```
name: DentTimeFeatureDriftHigh
expr: denttime_feature_psi > 0.25
for: 1m
labels:
  severity: warning
annotations:
  description: PSI of one monitored feature is above 0.25
  summary: DentTime feature drift is high
```

Labels	State	Active Since	Value
alertname=DentTimeFeatureDriftHigh feature=doctor_pct_long instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.278025090744663
alertname=DentTimeFeatureDriftHigh feature=appt_day_of_week instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	9.096452070416177
alertname=DentTimeFeatureDriftHigh feature=doctor_median_duration instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.3977903273596773
alertname=DentTimeFeatureDriftHigh feature=tooth_count instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	12.810405162786331
alertname=DentTimeFeatureDriftHigh feature=clinic_median_duration instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.535323109434887
alertname=DentTimeFeatureDriftHigh feature=clinic_pct_long instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	12.443981642430938
alertname=DentTimeFeatureDriftHigh feature=is_first_case instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.3320822229504347
alertname=DentTimeFeatureDriftHigh feature=appointment_rank_in_day instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.974971053860303
alertname=DentTimeFeatureDriftHigh feature=has_tooth_no instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	10.39775983516138
alertname=DentTimeFeatureDriftHigh feature=treatment_class instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	4.79744066553376
alertname=DentTimeFeatureDriftHigh feature=has_dentist_id instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	7.261768771657267
alertname=DentTimeFeatureDriftHigh feature=surface_count instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.616947499100291
alertname=DentTimeFeatureDriftHigh feature=appt_hour_bucket instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	5.99343876619051
alertname=DentTimeFeatureDriftHigh feature=total_amount instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	13.40453116288877
alertname=DentTimeFeatureDriftHigh feature=is_area_treatment instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	1.4691007098543662

DentTimeMacroF1Drop (0 active)

```
name: DentTimeMacroF1Drop
expr: denttime_macro_f1 < (denttime_macro_f1_baseline - 0.05)
for: 1m
labels:
  severity: critical
annotations:
  description: Current macro F1 is more than 0.05 below baseline
  summary: DentTime macro F1 dropped
```

DentTimeUnderEstimationHigh (0 active)

```
name: DentTimeUnderEstimationHigh
expr: denttime_underestimation_rate > (denttime_underestimation_rate_baseline + 0.05)
for: 1m
labels:
  severity: critical
annotations:
  description: Risk of scheduling too short is increasing
  summary: DentTime under-estimation rate increased
```

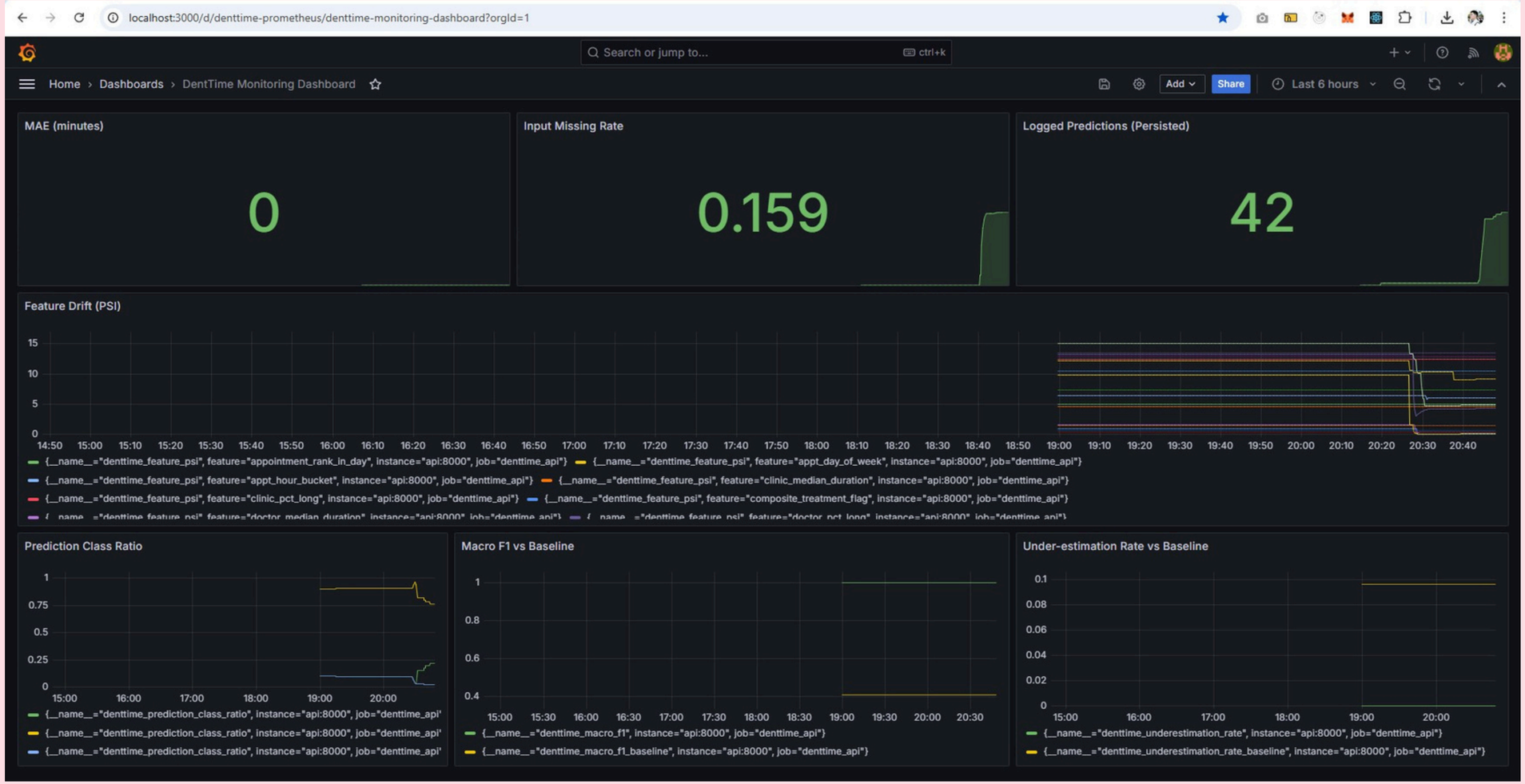
DentTimeMissingRateHigh (1 active)

```
name: DentTimeMissingRateHigh
expr: denttime_input_missing_rate > 0.1
for: 1m
labels:
  severity: warning
annotations:
  description: Input quality may be degrading
  summary: DentTime input missing rate is high
```

Labels	State	Active Since	Value
alertname=DentTimeMissingRateHigh instance=api:8000 job=denttime_api severity=warning	FIRING	2026-04-22T14:46:16.458246286Z	0.15897435897435896

PROMETHEUS ALERTS/FIRING

ACTUAL EVIDENCE FROM THE WORKING SYSTEM

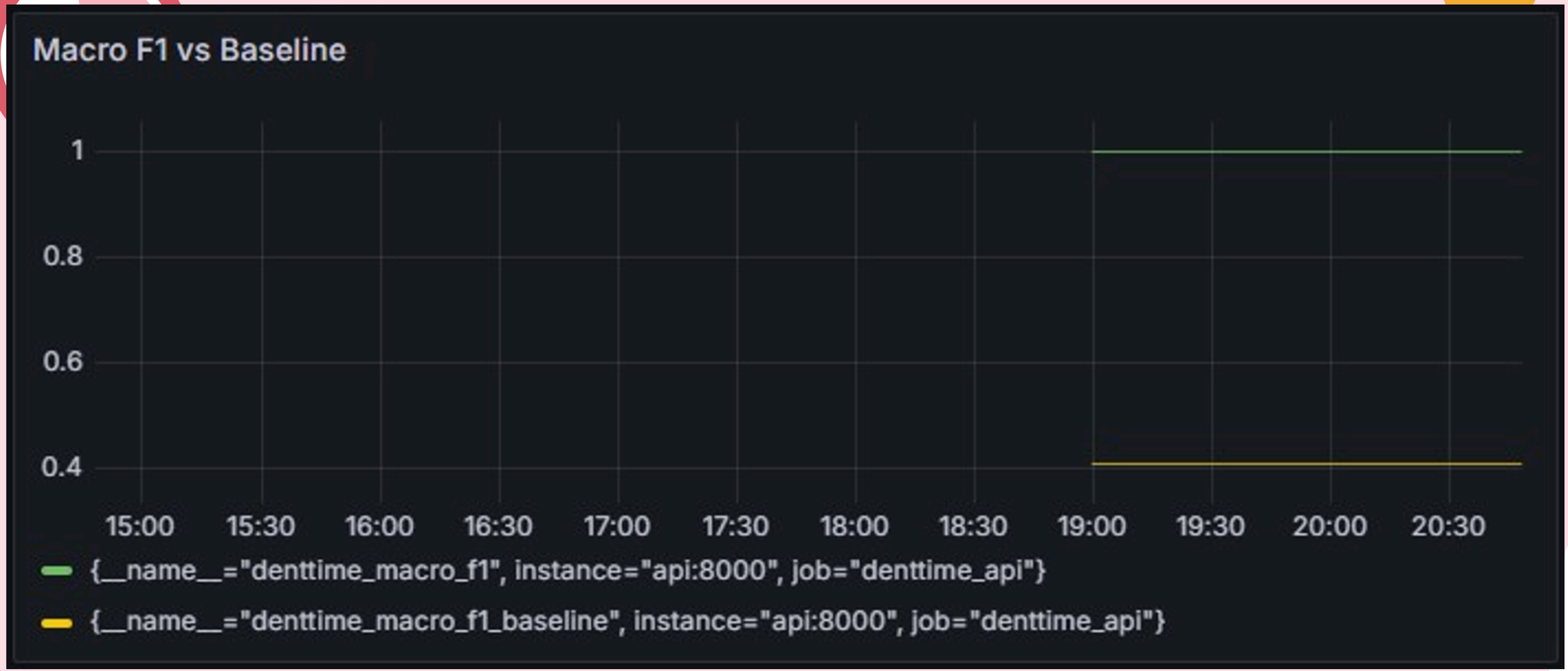


GRAFANA DASHBOARD ค่าจริง

HANDLING DEGRADATION AND OPERATIONAL ACTION

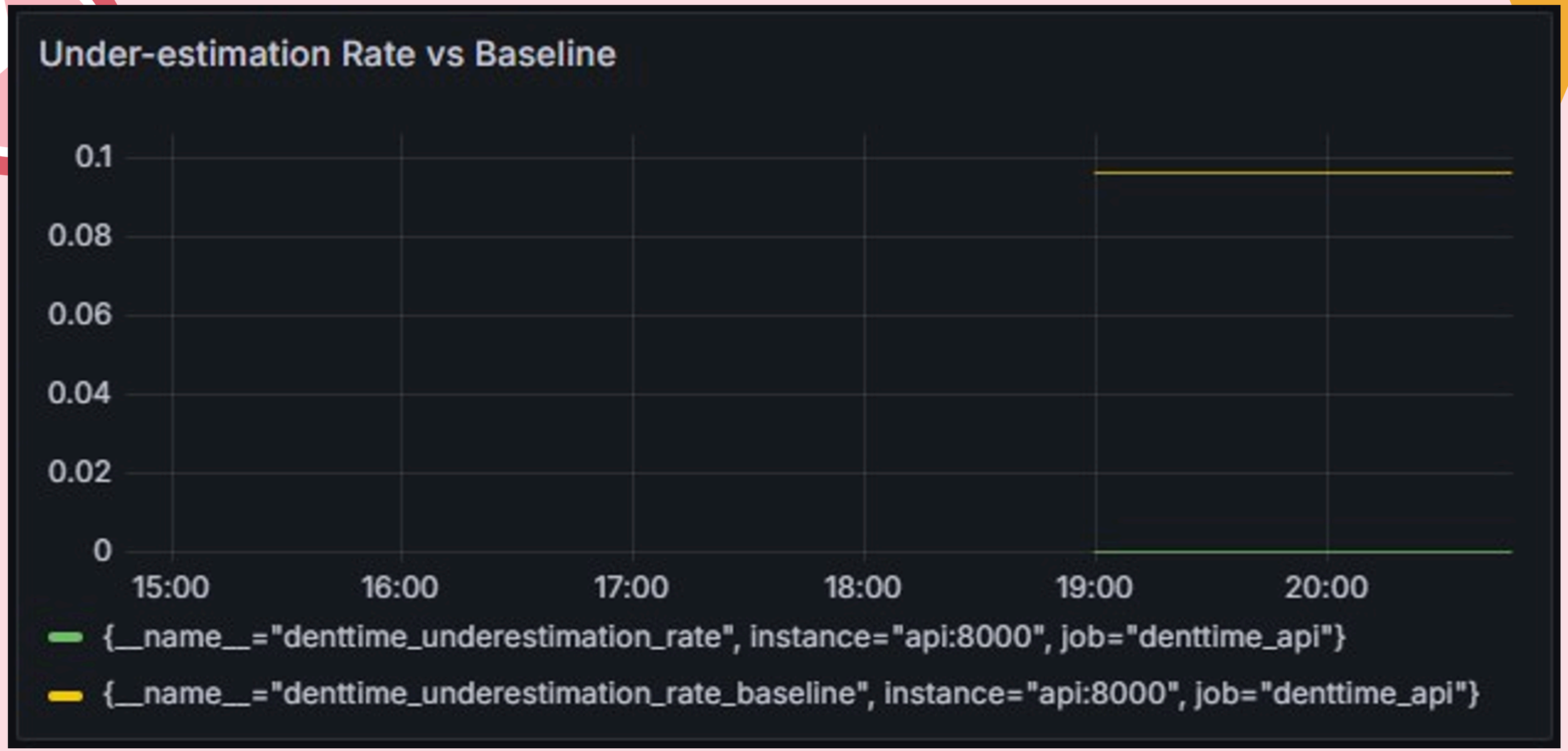
- If drift is high → inspect feature/source change
- If macro F1 drops vs baseline → retrain review
- If under-estimation rises → protect scheduling safety
- If input missing rate rises → validate upstream data/API contract
- Monitoring is tied to action, not only visualization

HANDLING DEGRADATION AND OPERATIONAL ACTION



GRAFANA PANEL “MACRO F1 VS BASELINE”

HANDLING DEGRADATION AND OPERATIONAL ACTION



GRAFANA PANEL “UNDER-ESTIMATION RATE VS BASELINE”

LIMITATIONS AND NEXT STEP

Current deployment is local/containerized prototype

Some alerts are rule-based and threshold-based

Monitoring depends on actual labels returning to the system

Next step: auto notification + retrain trigger + CI/CD integration